Marcel Torné Villasevil
marceltorne.github.io

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EDUCATION

- **Harvard University** — Cambridge, MA
  Master of Engineering in Computational Science and Engineering; GPA: 4.00/4.00 — *Aug. 2021 – May 2023*
  - Relevant courses : Robotic Manipulation (Prof. Russ Tedrake), Sensorimotor Learning (Prof. Pulkit Agrawal), Advances in Computer Vision (Prof. Philip Isola, Prof. Bill Freeman), Natural Language Processing (Prof. Jacob Andreas, Prof. Yoon Kim)
  - ***Cross-registered student at the Massachusetts Institute of Technology***
    - Relevant courses : Robotic Manipulation (Prof. Russ Tedrake), Sensorimotor Learning (Prof. Pulkit Agrawal), Advances in Computer Vision (Prof. Philip Isola, Prof. Bill Freeman), Natural Language Processing (Prof. Jacob Andreas, Prof. Yoon Kim)
- **École Polytechnique Fédérale de Lausanne, EPFL** — Lausanne, Switzerland
  Bachelor of Science in Computer Science; Grade: 5.62/6 (top < 3%) — *Sept. 2018 -- July. 2021*

CONFERENCE PUBLICATIONS

- ***Breadcrumbs to the Goal: Goal-Conditioned Exploration from Human-in-the-loop feedback***
  Marcel Torné Villasevil, *Max Balsells I Pamies, Zihan Wang, Samedh Desai, Tao Chen, Pulkit Agrawal, Abhishek Gupta*
  Neural Information Processing (NeurIPS) 2023 / arxiv / website
- ***Autonomous Robotic Reinforcement Learning with Asynchronous Human Feedback***
  Max Balsells I Pamies,* ***Marcel Torné Villasevil****, Zihan Wang, Samedh Desai, Pulkit Agrawal , Abhishek Gupta*
  Conference on Robot Learning (CoRL) 2023 / paper
  * indicates equal contribution

RESEARCH EXPERIENCE

- **Massachusetts Institute of Technology, CSAIL, Improbable AI Lab** — Cambridge, MA
  Graduate researcher supervised by Prof. Pulkit Agrawal and Prof. Abhishek Gupta — *Feb 2022 – Present*
  - My research is on leveraging Human Feedback to teach new skills to robots in the real world. I am tackling this problem in three different lines, first, designing new algorithms to incorporate crowdsourced noisy human feedback for policy learning, designing new algorithms for learning directly in the real world, and working on new ways to close the sim to real gap so that we can transfer the skills learned in the simulator from crowdsourced demonstrations to the real world.
- **Microsoft** — Bellevue, WA
  Data Scientist Intern — *June 2022 - August 2022*
  - **Microsoft Bing:** Built the first implementation of Extreme Classification algorithms by Microsoft Research in the Image ranker of Bing.
- **EPFL - Machine Learning and Optimization Lab (MLO)** — Lausanne, Switzerland
  Bachelor's Thesis research supervised by Prof. Martin Jaggi — *Feb 2021 - July 2021*
  - Developed a mobile application in JavaScript for privacy-preserving decentralized Machine Learning in medical image classification. [https://github.com/epfml/disco]
- **EPFL - Formula Student Racing Team**
  Autonomous driving team — *Feb 2021 - July 2021*
  - At the EPFL Racing Team, I pursued an individual Machine Learning Research Project. More concretely, I developed an algorithm for the Control and Planning of an autonomous vehicle using Deep Reinforcement Learning.

SELECTED HONORS

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- **Harvard Real Colegio Complutense Scholarship** *Aug. 2021 - May 2023*
  Merit-based scholarship for Spanish students studying at Harvard. Covers around 70% of the Master's tuition.
- **Harvard IACS Master Thesis proposal award** *Aug. 2021 - May 2023*
  Award given to a small, selected number of students in the master's program at Harvard for their excellent Master's thesis proposal. Covers around 10% of the total master's tuition.
- **Nova 111 Student List 2022 Spain** *May 2022*
  I was selected as one of the top 11 best students from Spain in Computer Science aged between 18-25
- **Excellence in the University Entrance Exam (top < 1%)** *July 2018*
  Award given by the Catalan president himself
- **Bronze Medal Spanish National Physics Olympiad 2018** *June 2018*
- **City of Reus High School Academic Excellence Award** *September 2016*
  Price given to the best student of each high school by the mayor

TEACHING EXPERIENCE

- **École Polytechnique Fédérale de Lausanne, EPFL** Lausanne, Switzerland
  Teaching Assistant in Computer Architecture *Sept. 2020 – Jan. 2021*
  - Lead a team of a dozen highly skilled graduate students. I owned the major project of the course, programming the Game of Life by John Horton Conway in assembly code.
- **École Polytechnique Fédérale de Lausanne, EPFL** Lausanne, Switzerland
  Teaching Assistant in General Physics I: Mechanics *Sept. 2019 - Dec. 2019*
  - Weekly meetings with a group of 20-40 students, I assisted reviewing lectures and helping students out with problem sets.

INDUSTRY EXPERIENCE

- **Amazon** Barcelona, Spain
  Software Engineering Intern *May 2020 - September 2020*
  - **Amazon Search, Science and AI:** *Developed browser plugin tool using Angular JS to help ML engineers debug their ranking algorithms. Tool used by more than 100 times per day by more than 20 engineers.*

LEADERSHIP EXPERIENCE

- **Quiz.Fit** Lausanne, Switzerland
  Co-founder and main developer *June 2019 – January 2021*
  - *Developed a mobile application in Swift for improving users' health. Convinced early investor, a professional in health insurance.*

OUTREACH AND SERVICE

- **Volunteering for Chromosomic Disorders**
  I am particularly concerned about chromosomal disorders. I have participated in multiple events to raise awareness, collect money for research, and better integration of these individuals into society.
  - **Regatta Carla for the RETT syndrome 2018**

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Helped with the social media and promotion of a sailing regatta to raise funds for research on the RETT syndrome.

- **Nastic Genuine 2018**
  Volunteered in the organization of a Spanish National football tournament for people with Down syndrome.
- **La Marato de TV3 2012-2018**
  On multiple occasions, I performed in music, both playing the baroque guitar and singing, to collect money for multiple causes, one of which was collecting money towards research on chromosomic disorders.

- **Mentorship for students 2018 - Present**
  Other things I care about is helping students from Spain achieve their maximal academic potential, mentor students to get into top programs in Europe and the US, and mentor them on career path.
  - **Google Developer Student Club of Lausanne November 2022**
    Gave a talk explaining my experience during my internship at Microsoft and the whole process of job finding in the United States.

Other Activities

- **Running** — Cambridge, MA
  2022 - Present
  I started running seriously very recently, and it has become a big part of my weekly routine to train my mind and body.
  - **Nice Ironman 70.3 2023** — *26th June 2023*
    I ran a half-ironman in the city of Nice. This is one of the toughest triathlon races there are, consisting in a total of 70.3 miles between swimming, biking and running, and I completed it in under 6hours 43 minutes.
  - **Boston Half Marathon 2022** — *13th November 2022*
    The BBA, Boston Half Marathon was my first running race ever. I am proud of my performance having ran it in 1 hour 42 minutes.

- **Sailing** — Salou, Spain
  2008 - Present
  - **Windsurf World Championship 2017 & 2018**
    I competed at the highest level in windsurfing for three years. I was sailing in the Raceboard modality; this is a longer table for upwind and downwind sailing.
  - **29er National Championship 2015 & 2016**
    I competed at international level in 29er sailboat. This is a two-handed sailboat, considered one of the toughest boats to sail because of its instability.
  - **Optimist Catalan Championship 2010 – 2014**
    I competed at regional level in the optimist sailboat. This is a single-handed sailboat for kids up to the age of 16. This modality gets very competitive, and I participated at the international regatta of Riva del Garda, Italy, with up more than 1000 participants.

- **Undergraduate Music degree** — Vila-seca, Spain
  Conservatory of music — *2008 - 2018*
  Music used to be a very big part of my life, I did 10 years of music degree, which used to take from three to five afternoons per week depending on the semester. I specialized in Baroque Guitar and the Lute and learned from an internationally known musician in the field, William Waters. I played in Chamber Music accompanied by both voice and other string instruments, guitars, and violins.

Skills

- **Languages**

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- **Catalan -** *mother tongue*
- **Spanish -** *mother tongue.*
- **French -** *professional proficiency*

- **Programming Languages**
  - *Python, Javascript, C, C++, Scala, Spark, Kotlin SQL, HTML, CUDA, VHDL, MIPS assembly*
  - *ML libraries: Pytorch, JAX, TensorFlow, Pytorch Lightning, Weights and Biases*
  - *Robotics Simulators: Mujoco, Pybullet, Drake*